

POTHOLELOG · DESIGN PRECIS

PotholeLog design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Approved follow-ups carried out: light pipes and LEDs, breather vent and nameplate label as BOM lines 14 to 16; mass 0.44 kg and cost USD 77.50 (R15 not met); segment file pooling rule written into the server export
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02 (PHL-DEC-001): lid light pipes, nameplate and breather vent adopted; clear window renders only; CityTwin ingest and pooled dates; first pilot partner
0.5	2026-10-01	Amish Chadha	Constructable design (PHL-DDR-003): carrier plate, box fixing, component table and key numbers updated
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update (numbers checked against PHL-CAL-001, design choices adopted per PHL-DDR-001, parametric model and drawing PHL-DWG-001, detection and transient findings)

0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

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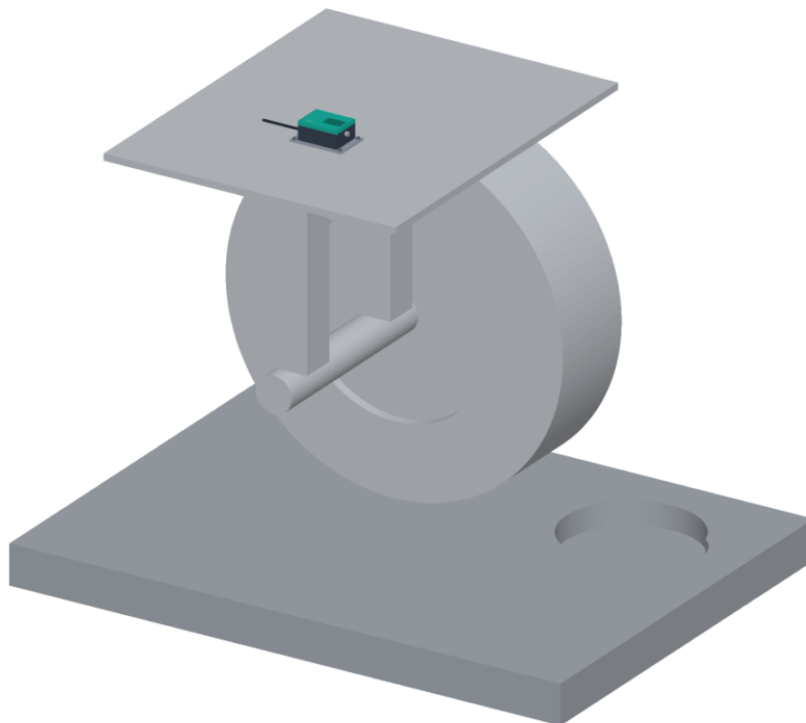
■ Summary

PotholeLog is a sealed box, 120 x 90 x 55 mm on a 4 mm plate (59 mm overall), bolted to the floor of a bus or refuse truck above the rear axle. It records how the vehicle shakes and where it is, turns each 100 m of road into a roughness value and flags sharp impacts as possible potholes, then uploads a small daily file over depot Wi-Fi. Averaged over many passes and calibrated per vehicle, the data gives a city a weekly map of road condition on every street its fleet drives. The TRL 3 calculations (PHL-CAL-001) show that off-the-shelf modules meet the sampling, storage, hold-up and power needs for an estimated \$77.50 in parts, \$2.50 over the \$75 value-engineering target since the status lights, breather vent and nameplate adopted on 2026-10-02 were added, and that the roughness signal is well above sensor noise. They also found two limits, both now addressed by Amish's 2026-09-25 decisions (PHL-DDR-002). A bus tyre bridges most of a 300 mm pothole, so the floor-mounted logger finds the R1 reference pothole only at low speed (up to about 20 km/h on a fair road, 50 km/h on a good one); small potholes are therefore sought on slow passes, 10 to 30 km/h, near stops and junctions. The converter is now rated 60 V so that it survives a load dump on a 24 V vehicle. Detection accuracy and calibration to IRI remain unverified until field data exist. On 2026-10-01 the design was made constructable (PHL-DDR-003): the modules now sit on a carrier plate held by the four hex standoffs that fix the box to the mounting plate, and the build is described component by component in the prototype build plan (PHL-BLD-001).

PotholeLog

github.com/BoujeeEnjinia1701/potholelog

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: Road with pothole, Wheel, axle, suspension and floor section for scale

Figure 1. Concept massing model. The logger is the small dark box on the floor section; the grey scene is for scale.

■ How it works

1. **Sense.** A 6-axis IMU rigidly fixed to the enclosure floor samples acceleration at 400 Hz. A GNSS module under the lid logs position and speed at 10 Hz.
2. **Log raw data.** Raw samples go to a microSD card as a rolling buffer, so any detection can be re-run later with a better algorithm.
3. **Summarize on board.** For every 100 m traveled, the controller computes a roughness value from the vertical acceleration (band-limited RMS from 0.5 to 20 Hz, tagged with its speed band for calibration), and records the segment's start and end, speed and the number of samples. Sharp vertical impacts above a threshold are stored as defect events, with most small potholes expected from slow passes at 10 to 30 km/h near stops and junctions (R5) with position, speed and peak size; the front and rear axle impacts of the same defect, about 0.4 to 0.7 s apart in town, are paired to confirm the event and place it at the rear axle.
4. **Upload.** When the vehicle is back within depot Wi-Fi with the ignition on, on arrival or at the next start, the controller joins the network and uploads the day's summaries (about 112 kB, about 8 s) to the fleet's server. The hold-up capacitor cannot power Wi-Fi long enough to upload after the ignition is switched off.
5. **Calibrate and aggregate.** On the server, each vehicle's raw roughness is converted to an IRI estimate using a calibration equation found on reference sections, in the way World Bank Technical Paper 46 describes for response-type systems (Sayers et al., 1986). Events from several passes and vehicles are clustered; a cluster seen on repeated passes becomes a reported defect, as in Pothole Patrol (Eriksson et al., 2008).
6. **Publish.** The city publishes a map of segment roughness and defect clusters as CSV and GeoJSON, for its own GIS and for CityTwin. Vehicle tracks and timestamps stay on the operator's server. CityTwin fetches the segment files from the operator's server; a date per segment is published only once at least two vehicles or several days are pooled, and pass times never are (decided 2026-10-02, PHL-DEC-001). The server export applies the pooling rule below.

PotholeLog: data flow (values from PHL-CAL-001; road data only, no images or audio)
CONCEPT, NOT FOR FABRICATION

github.com/BoujeeEnjinia1791/potholelog



Figure 2. Data flow. Values from PHL-CAL-001.

Server export: segment files and the pooling rule

The operator's server writes one segment file a day in CSV and GeoJSON, following the CityTwin open data export, and CityTwin fetches it over HTTPS (decided 2026-10-02, PHL-DEC-001; CityTwin's outbound pull, CTW-DDR-001 D11). Each row is one 100 m segment: segment identifier and geometry, calibrated roughness (IRI estimate) with its uncertainty, number of passes, number of vehicles and number of days pooled, and defect clusters as point events. The pooling rule is applied in the export, not left to the reader:

1. A segment's row carries a date (day only, never a time) only when its value pools passes from at least two vehicles, or from at least three separate days. Otherwise the date field is left empty.

- 2. Pass times, vehicle identifiers, tracks and speeds are never exported; they stay on the operator's server.
- 3. Defect clusters carry the date of the newest day in the pool, under the same rule.

The three-day figure is this precis's reading of "several days" in the decision; it is to be matched with CityTwin's item 6 so both repos use the same rule and field names.

■ Main components

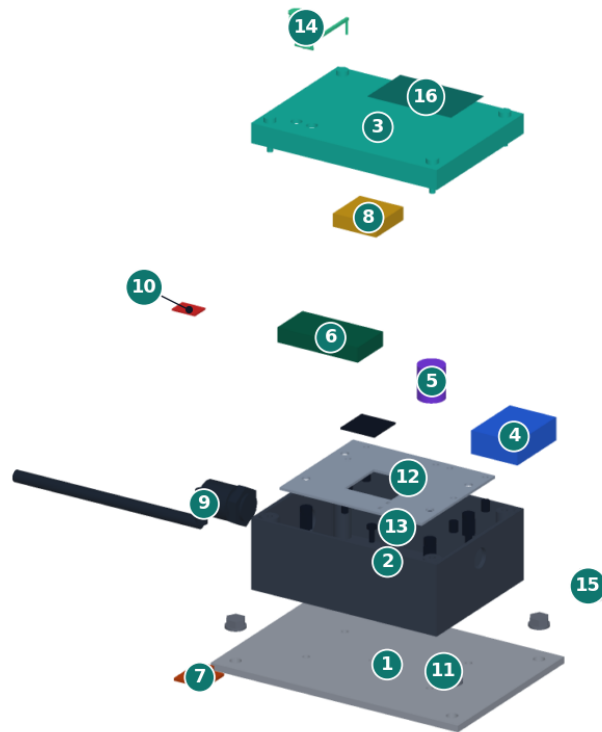
Table 1. Main components. Numbers match the exploded view (Figure 3), `cad/src/model.py` and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	Mounting plate	160 x 110 x 4 mm aluminum, 4 x M6 on 140 x 90 mm; tapped for the box standoffs and the IMU screws	Bolts to existing crossmember, seat-rail or floor fixings; first mode about 168 Hz on corner bolts
2	Enclosure base	Stock IP65 ABS or polycarbonate box, 120 x 90 mm, with corner lid-screw pillars	Held to the plate by four M4 hex standoffs through its floor (PHL-DDR-003); a 12.2 mm hole in the front end wall for the breather vent (15)
3	Enclosure lid	Supplied with the box, with gasket; opaque; two 6.4 mm holes for the status light pipes (14) and the nameplate label (16) on top (decided 2026-10-02)	GNSS antenna under the lid; plastic lid keeps sky view through vehicle windows; the clear window shown in the renders is for the renders only
4	DC-DC converter and protection	9 to 60 V in (60 V rated), 5 V 1 A out; TVS diode, reverse-polarity diode, input fuse	Covers normal 12 V and 24 V supplies and a 24 V suppressed load dump (58 V), clearing the TVS clamp by 1.9 V (PHL-DDR-002, N1)
5	Hold-up supercapacitor	1 F, 5.5 V, rated to 70 °C or more	About 7 s at end of life to close files when power drops
6	Controller	ESP32-S3 board with Wi-Fi and microSD slot; module variant rated to 85 °C	Decided by Amish, 2026-09-25 (PHL-DDR-001, D4)
7	IMU	6-axis, LSM6DSO class, ± 16 g, 400 Hz or more	On the box floor, held by two M3 screws through the floor into the plate; gyroscope helps separate body roll and pitch from vertical motion
8	GNSS module	u-blox M10 class with patch antenna, 10 Hz, PPS output	Under the lid on a foam tape pad; PPS time-stamps the IMU samples
9	Cable gland and fused lead	M16 gland, 2 m lead, inline 2 A fuse	Wired to an ignition-switched fused circuit
10	microSD card	32 GB high-endurance, -25 to 85 °C	133 to 160 days of raw data
11	Fixings	4 x M6 (8.8) with locking nuts and washers	Crash load is small; loosening needs a later test
12	Module carrier plate	92 x 76 x 1.5 mm aluminum with a window over the IMU	Carries the converter, supercapacitor and controller (PHL-DDR-003)
13	Box fixing kit	4 x M4 hex standoffs, screws, nylon standoffs, foam tape, sealant	Added for construction (PHL-DDR-003)
14	Status light pipes and LEDs	Two sealed panel-mount light pipes in the lid, a 5 mm LED in each, plug lead to the controller	Power and logging lights; adopted 2026-10-02 (PHL-DEC-001, decision 3)
15	Breather vent	M12 pressure-equalizing membrane vent, IP67, in the front end wall	Stops the box pumping moist air through the gasket as it heats and cools; adopted 2026-10-02 (decision 4)
16	Nameplate label	54 x 34 mm laminated polyester label with a forward arrow	Shows which way the IMU's X axis must point; adopted 2026-10-02

PotholeLog: exploded viewgithub.com/BoujeeEnjinia1701/potholelog

CONCEPT, NOT FOR FABRICATION

- 1 Mounting plate, aluminum
- 2 Enclosure base, IP65
- 3 Enclosure lid with gasket
- 4 DC-DC converter and protection
- 5 Hold-up supercapacitor
- 6 Controller, Wi-Fi, microSD slot
- 7 6-axis IMU
- 8 GNSS module and patch antenna
- 9 Cable gland and fused lead
- 10 microSD card, 32 GB
- 11 M6 bolts, washers, locking nuts
- 12 Module carrier plate
- 13 Hex standoffs, screws, foam tape
- 14 Status light pipes and LEDs
- 15 Breather vent, M12
- 16 Nameplate label



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 3. Exploded view with numbered callouts matching the BOM.

PotholeLog: cutaway

github.com/BoujeeEnjinia1701/potholelog

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 4. Cutaway on the plane through the IMU. The IMU sits on the enclosure floor, the controller, supercapacitor and converter behind it, and the GNSS module under the lid.

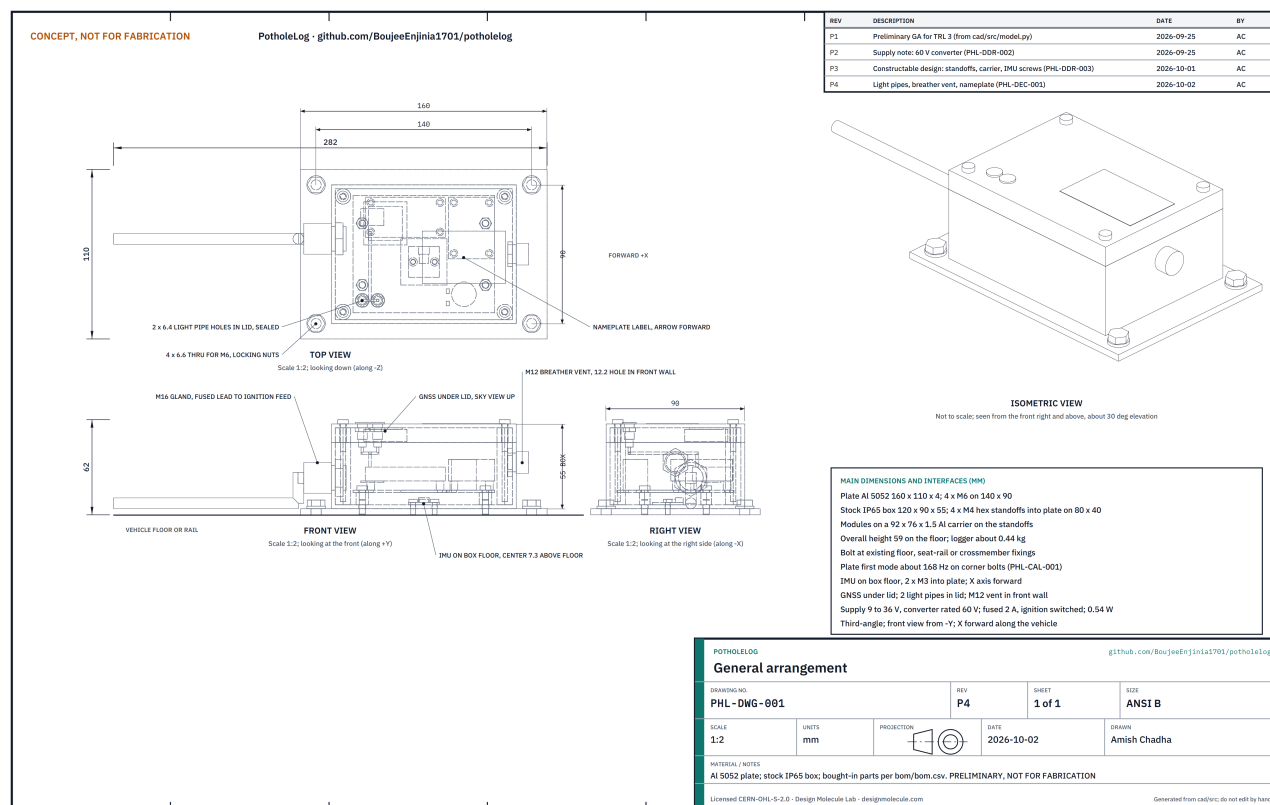


Figure 5. General arrangement PHL-DWG-001, Rev P3, generated from `cad/src/model.py` (PDF). Preliminary, not for fabrication.

■ Key numbers

All values are from PHL-CAL-001 v0.3 (`docs/04-calcs/sizing.py`), which states the assumptions; they are calculations for review, not measurements.

Table 2. Key numbers.

Quantity	Value	Basis	Requirement
Sample spacing on the road	35 mm at 50 km/h, 56 mm at 80 km/h	400 Hz	R6
Samples across a 300 mm pothole	9 at 50 km/h, 5 at 80 km/h	Length divided by spacing	R1
Tyre drop into a 300 mm pothole	22 mm, whatever its depth	1.04 m tyre, rigid-circle envelope	R1
Pothole detection margin	Above 1 up to about 20 km/h (IRI 4) and 50 km/h (IRI 2); in the 10 to 30 km/h band, 1.94 or more (IRI 2) and 0.97 at 30 km/h (IRI 4)	Quarter-car model against roughness crossing once per km	R1, R5 at risk
Roughness signal against noise	12 times at 20 km/h on a smooth road	0.5 to 20 Hz band	R5
Synthetic calibration	r^2 0.93 from one pass, 0.99 from five	40 sections, IRI 1 to 8 m/km	R3
Location, 95 %	2.3 m suburban, 18.2 m dense urban (10 passes)	M10-class GNSS with PPS	R4 at risk
Raw data	5.23 kB/s; 188 MB per 10 h day	6 axes x 16 bit at 400 Hz, plus GNSS and time stamps	
Raw storage on a 32 GB card	133 to 160 days	10 to 12 h of driving a day	R7 met
Summaries per day	112 kB, uploaded in about 8 s	200 km, 48 B segments, 500 events at 32 B	R8 met
Running power	0.54 W; 1.42 W for about 8 s while uploading	Loads 91 mA at 3.3 V, linear regulator, 85 % converter	R9
Supply current	45 mA at 12 V, 22 mA at 24 V	From running power	
Hold-up time	10.0 s new, 7.0 s at end of life	1 F from 4.7 V to 3.6 V	R10 met; 1 s needed
Inside temperature	About 72 °C at +70 °C ambient	2.0 K self-heating	R11 at risk
Plate first mode	168 Hz on corner bolts	4 mm aluminum, 140 mm span	R12 met on paper
Installation	30 min	Task estimate	R12 at the limit
Size	120 x 90 x 55 mm box on a 160 x 110 x 4 mm plate, 59 mm overall	<code>cad/src/model.py</code>	
Mass	0.44 kg; 0.58 kg with lead and fixings	Model volumes and module masses	
Converter input rating	60 V against a 58 V suppressed load dump and a 58.1 V TVS clamp	ISO 16750-2 test B levels, to confirm	R9 at risk (TVS pulse energy)

Quantity	Value	Basis	Requirement
Parts cost	\$77.50	bom/bom.csv	R15 not met: \$2.50 over the \$75 value-engineering target

■ Key design choices

The choices below were proposed at TRL 2 and decided by Amish on 2026-09-25 (go with recommendation; PHL-DDR-001 and PHL-DDR-002).

- **Sprung-mass mount on the floor, not on the axle.** The floor is clean, dry and easy to reach, and the GNSS can see the sky through the windows. The axle would give a stronger, less filtered signal but sees large shocks, water and stones. Calibration per vehicle corrects for the suspension. PHL-CAL-001 supports the choice: the axle would see about 9 g, 65 times the floor peak.
- **Depot Wi-Fi upload, not cellular.** It keeps parts cost near the value-engineering target and avoids data plans. Cellular (for example an LTE-M module) would give same-day data but adds about \$20 to \$30 per unit and a monthly fee; it stays an optional add-on, costed separately.
- **Raw data kept on the card.** Storage is cheap, and raw data lets the detection method improve without new hardware.
- **Summaries and events, not tracks, leave the vehicle's operator.** Only road-level results are published (R13).
- **Controller, IMU and GNSS.** ESP32-S3 (Wi-Fi, low cost, wide community), an LSM6DSO-class IMU and a u-blox M10-class GNSS. The alternatives considered were an RP2040 board with a separate Wi-Fi module and an nRF52 board with a phone or gateway for upload.
- **100 m segments and point events, in CSV and GeoJSON** following the CityTwin open data export.
- **60 V converter input.** A 60 V-rated converter replaces the 36 V part so that a 24 V suppressed load dump does not destroy it; about \$2 more, with `budget_usd` raised from \$70 to \$75 (PHL-DDR-002, N1). A surge-stopper front end would be more robust and was not chosen.
- **Slow-pass defect detection.** The R5 defect-detection range is 10 to 30 km/h, with the 300 mm reference defect kept, because buses slow near stops and junctions and pass each spot many times (PHL-DDR-002, N2).
- **Bicycles later.** This build serves buses and refuse trucks; a bicycle variant with its own battery and mount is a later step, and the pitch is unchanged.
- **Built on existing lab work where it fits.** Outputs follow the CityTwin open data export so PotholeLog segments can appear on the CityTwin map; CityTwin is built on TwinKit. PotholeLog does not use FieldNode, because it runs from vehicle power rather than solar and uploads at the depot rather than over LoRaWAN.

■ Safety

Safety: PotholeLog is fitted to vehicles that carry passengers and operate in traffic.

- **Secure mounting.** A loose box can become a projectile in a crash or hard stop. Bolt the plate through with M6 bolts and locking nuts at existing fixing points, never with adhesive or magnets alone. Do not cut, drill or weld chassis members or safety-critical structure; follow the fleet operator's and vehicle maker's rules for body fittings.

- **Vehicle electrics.** Connect only to a fused, ignition-switched circuit through the inline fuse, with the battery isolated during installation. Route the lead away from moving parts, hot surfaces and sharp edges, and protect it where it passes through panels. A short on a 24 V vehicle supply can start a fire. The converter is now rated 60 V for a 24 V suppressed load dump (PHL-DDR-002), but the margin over the TVS clamp is only 1.9 V and the TVS pulse energy is untested; do not fit a converter rated below 60 V to a 24 V vehicle.
- **Working on vehicles.** Install only with the vehicle parked, secured and switched off. Never work under a vehicle supported only by a jack.
- **Driver distraction.** The logger has no controls or display and must not be handled while the vehicle is moving.
- **Sharp edges.** Deburr the aluminum plate.
- **Supercapacitor.** It stays charged for a short time after power is removed; do not short its terminals.
- **Data.** Vehicle location data can reveal where drivers are and when. Keep tracks on the operator's server under the operator's data policy, inform drivers and unions before a pilot, and publish only road-level results.

This is a research prototype. It is not certified to automotive electrical or EMC standards, and its outputs must not be used for contract acceptance or safety decisions without validation against a calibrated reference.

■ Open questions

- Does the TVS survive a 24 V load dump pulse with the 60 V converter, or is a surge stopper needed (R9)? Needs a bench test (TRL 4, on hold).
- Are slow passes near stops and junctions frequent enough to meet R1 with the 10 to 30 km/h detection band (R5)? Needs field data.
- How far off is GNSS in the partner city's densest streets, and is map matching needed for R4?
- Which host fleet and city for a first pilot, and who provides reference IRI? Decided 2026-10-02 (PHL-DEC-001): the first candidate to approach is a Dallas-Fort Worth area city with its own bus or refuse fleet, with TxDOT pavement management roughness data on state roads in that city as the reference IRI sections.
- Should a bicycle variant be developed later, and with what battery and mount (R16)?

Concept media: [blueprint sheet](#), [interactive 3D model](#). Calculations: [PHL-CAL-001](#). Decisions: [PHL-DDR-001](#).